

Quick start for LaTeXing with IEEEtran.cls for IEEE Computer Society Conferences

First Author, Second Author
University of Examples, Germany
{lastname}@example.org

Third Author
School of Electrical and
Computer Examples
Georgia Institute of Examples
Atlanta, Georgia 30332-0250
<http://www.example.org>

Abstract—Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

I. INTRODUCTION

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus

viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus enim. Vestibulum pellentesque felis eu massa. TODO!

The remainder of the paper starts with a presentation of related work (Section II). It is followed by a presentation of hints on L^AT_EX (Section III). Finally, a conclusion is drawn and outlook on future work is made (Section IV).

II. RELATED WORK

Winery [1] is a graphical modeling tool. The whole idea of TOSCA is explained by Binz et al. [2].

III. L^AT_EX HINTS

This section contains hints on writing L^AT_EX. It focuses on minimal examples, which can be directly adapted to the content

A. Handling of paragraphs

One sentence per line. This rule is important for the usage of version control systems. A new line is generated with a blank line. As you would do in Word: New paragraphs are generated by pressing enter. In L^AT_EX, this does not lead to a new paragraph as L^AT_EX joins subsequent lines. In case you want a new paragraph, just press enter twice! This leads to an empty line. In word, there is the functionality to press shift and enter. This leads to a hard line break. The text starts at the beginning of a new line. In L^AT_EX, you can do that by using two backslashes (\\). This is rarely used.

Please do *not* use two backslashes for new paragraphs. For instance, this sentence belongs to the same paragraph, whereas the last one started a new one. A long motivation for that is provided at <http://loopspace.mathforge.org/HowDidIDoThat/TeX/VCS/#section.3>.

Corresponding L^AT_EX code of ./paper.tex

```
577 %EüÜöÜ
578 One sentence per line.
579 This rule is important for the usage of version
    control systems.
580 A new line is generated with a blank line.
581 As you would do in Word:
582 New paragraphs are generated by pressing enter.
583 In LaTeX, this does not lead to a new paragraph
    as LaTeX joins subsequent lines.
584 In case you want a new paragraph, just press
    enter twice!
585 This leads to an empty line.
586 In word, there is the functionality to press
    shift and enter.
587 This leads to a hard line break.
588 The text starts at the beginning of a new line.
589 In LaTeX, you can do that by using two
    backslashes (\textbackslash\textbackslash).
590 \\
591 This is rarely used.
592
593 Please do \textit{not} use two backslashes for
    new paragraphs.
594 For instance, this sentence belongs to the same
    paragraph, whereas the last one started a
    new one.
595 A long motivation for that is provided at
    \url{http://loopspace.mathforge.org/HowDidIDoThat/T
```

B. Notes separated from the text

The package mindflow enables writing down notes and annotations in a way so that they are separated from the main text.

This is a small note.

Corresponding L^AT_EX code of ./paper.tex

```
602 %EüÜöÜ
603 \begin{mindflow}
604 This is a small note.
605 \end{mindflow}
```

C. Handling TODOs

Markierter Text.

Corresponding L^AT_EX code of ./paper.tex

```
610 %EüÜöÜ
611 \textmarker{Markierter Text.}
```

Bei `\textmarker` wird nur die Textfarbe geändert, da dies auch bei einigen Worten gut funktioniert.

Markierter Text.

Corresponding L^AT_EX code of ./paper.tex

```
616 %EüÜöÜ
617 \textcomment{Markierter Text.}{Kommentar dazu.}
```

Manuelle Markierung für Text, der seit der letzten Version geändert wurde.

Corresponding L^AT_EX code of ./paper.tex

```
620 %EüÜöÜ
621 \modified{Manuelle Markierung für Text, der
    seit der letzten Version geändert wurde.}
```

Das ist ein Text. Geänderter Text.

Corresponding L^AT_EX code of ./paper.tex

```
624 %EüÜöÜ
625 Das ist ein Text.
626 \change{FL1: Text angepasst}{Geänderter Text}.
```

Hier nur ein Kommentar.

Corresponding L^AT_EX code of ./paper.tex

```
629 %EüÜöÜ
630 Hier nur ein Kommentar\sidecomment{Kommentar}.
```

TODO!

Corresponding L^AT_EX code of ./paper.tex

```
633 %EüÜöÜ
634 \todo{Hier muss noch kräftig Text produziert
    werden}
```

D. Hyphenation

L^AT_EX automatically hyphenates words. When using microtype, there should be fewer hyphenations than in other settings. It might be necessary to tweak the hyphenations nevertheless. Here are some hints:

In case you write “application-specific”, then the word will only be hyphenated at the dash. You can also write `applica\allowbreak{tion-specific}` (result: application-specific), but this is much more effort.

You can now write words containing hyphens which are hyphenated at other places in the word. For instance, `application"=specific` gets application"=specific. This is enabled by an additional configuration of the babel package.

Corresponding L^AT_EX code of ./paper.tex

```

644 %\begin{figure}
645 In case you write
        \enquote{application-specific}, then the
        word will only be hyphenated at the dash.
646 You can also write
        \verb!applica\allowbreak{}tion-specific!
        (result:
        applica\allowbreak{}tion-specific), but
        this is much more effort.
647
648 You can now write words containing hyphens
        which are hyphenated at other places in the
        word.
649 For instance, \verb!application="specific! gets
        application="specific.
650 This is enabled by an additional configuration
        of the label package.

```

E. Typesetting Units

Numbers can be written plain text (such as 100), by using the siunitx package as follows: $100 \frac{\text{km}}{\text{h}}$, or by using plain L^AT_EX (and math mode): $100 \frac{\text{km}}{\text{h}}$.

Corresponding L^AT_EX code of ./paper.tex

```

655 %\begin{figure}
656 Numbers can be written plain text (such as 100
        ), by using the
        \href{https://ctan.org/pkg/siunitx}{siunitx}
        package as follows:
657 \SI{100}{\km\per\hour},
658 or by using plain LATEX (and math mode):
659 $100 \frac{\mathit{km}}{\mathit{h}}$.

```

5 % of 10 kg

Corresponding L^AT_EX code of ./paper.tex

```

662 %\begin{figure}
663 \SI{5}{\percent} of \SI{10}{kg}

```

Numbers are automatically grouped: 123 456.

Corresponding L^AT_EX code of ./paper.tex

```

666 %\begin{figure}
667 Numbers are automatically grouped: \num{123456}.

```

F. Surrounding Text by Quotes

Please use the “enquote command” to quote something. Quoting with “quote” or “quote” also works.

Corresponding L^AT_EX code of ./paper.tex

```

672 %\begin{figure}
673 Please use the \enquote{enquote command} to
        quote something.
674 Quoting with "`quote" or ``quote" also works.

```

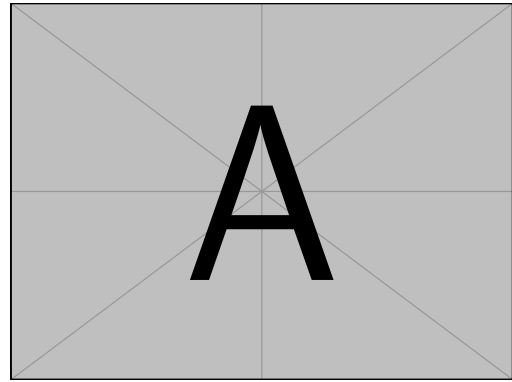


Figure 1. Example figure for cref demo

Heading1	Heading2
One	Two
Thee	Four

Figure 2. Example table for cref demo

G. Cleveref examples

Cleveref demonstration: Cref at beginning of sentence, cref in all other cases.

Figure 1 shows a simple fact, although Figure 1 could also show something else.

Figure 2 shows a simple fact, although Figure 2 could also show something else.

Section III-G shows a simple fact, although Section III-G could also show something else.

Corresponding L^AT_EX code of ./paper.tex

```

704 %\begin{figure}
705 \Cref{fig:ex:cref} shows a simple fact,
        although \cref{fig:ex:cref} could also show
        something else.
706
707 \Cref{tab:ex:cref} shows a simple fact,
        although \cref{tab:ex:cref} could also show
        something else.
708
709 \Cref{sec:ex:cref} shows a simple fact,
        although \cref{sec:ex:cref} could also show
        something else.

```

H. Figures

Figure 3 shows something interesting.

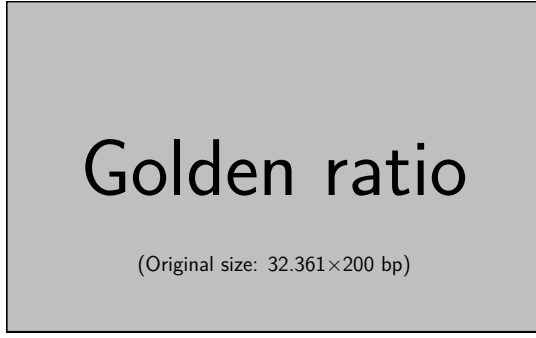


Figure 3. Simple Figure. Based on Scharrer [3].

Corresponding L^AT_EX code of ./paper.tex

```

714 %\begin{figure}
715 \Cref{fig:label} shows something interesting.
716
717 \begin{figure}
718   \centering
719   \includegraphics[width=.8\linewidth]{example-image-golden}
720   \caption[Simple Figure]{
721     Simple Figure.
722     Based on \citet{mwe}.
723   }
724   \label{fig:label}
725 \end{figure}

```

One can span a figure across multiple columns by using `\begin{figure*}`. See Figure 4 as an example.

Corresponding L^AT_EX code of ./paper.tex

```

731 %\begin{figure*}
732 \begin{figure*}
733   \centering
734   % note that \textwidth is used instead of
735   % \linewidth
736   % This ensures that the graphics width is 60%
737   % of the "page" (text block), and not just
738   % 60% of the current text column
739   % See
740   % https://tex.stackexchange.com/a/17085/9075
741   % for details
742   \includegraphics[width=.6\textwidth]{example-image-golden}
743   \caption[16x9 Figure]{
744     16x9 Figure.
745     Based on \citet{mwe}.
746   }
747   \label{fig:16x9}
748 \end{figure*}

```

I. Sub Figures

An example of two sub figures is shown in Figure 5.

Corresponding L^AT_EX code of ./paper.tex

```

748 %\begin{figure}
749 \begin{figure}
750   \centering
751   \subfloat[Case I]{\includegraphics[width=.4\linewidth]{example-image-a}}
752   \label{fig:first_case}}
753   \hfil
754   \subfloat[Case II]{\includegraphics[width=.4\linewidth]{example-image-b}}
755   \label{fig:second_case}}
756   \caption[Example figure with two sub figures.]{
757     Example figure with two sub figures.
758     IEEE style. (a) The first case. (b) The
759     second case.}
759   \label{fig:two_sub_figures}
760 \end{figure}

```

Note that often IEEE papers with subfigures do not employ subfigure captions (using the optional argument to `\subfloat[]`), but instead will reference/describe all of them (a), (b), etc., within the main caption. Be aware that for `subfig.sty` to generate the (a), (b), etc., subfigure labels, the optional argument to `\subfloat` must be present. If a subcaption is not desired, just leave its contents blank, e.g., `\subfloat[]`. An example is shown in Figure 6.

Corresponding L^AT_EX code of ./paper.tex

```

770 %\begin{figure}
771 \begin{figure}
772   \centering
773   \subfloat[]{\includegraphics[width=.4\linewidth]{example-image-a}}
774   \label{fig:first_case_ieee}}
775   \hfil
776   \subfloat[]{\includegraphics[width=.4\linewidth]{example-image-b}}
777   \label{fig:second_case_ieee}}
778   \caption[Example figure with two sub figures.]{
779     Example figure with two sub figures.
780     IEEE style. (a) The first case. (b) The
781     second case.}
781   \label{fig:two_sub_figures_ieee}
782 \end{figure}

```

J. Tables

Note that IEEE does not support `\begin{table}`, one has to use `\begin{figure}`.

Corresponding L^AT_EX code of ./paper.tex

```

787 %\begin{table}
788 \begin{figure}
789   \caption[Simple Table]{
790     Simple Table.
791     Based on \citet{mwe}.
792   }
793   \centering
794   \begin{tabular}{ll}
795     \toprule
796     Heading1 & Heading2 \\
797     \midrule
798     One      & Two      \\
799     Three    & Four     \\
800     \bottomrule
801   \end{tabular}
802   \label{fig:simple_table}
803 \end{figure}

```

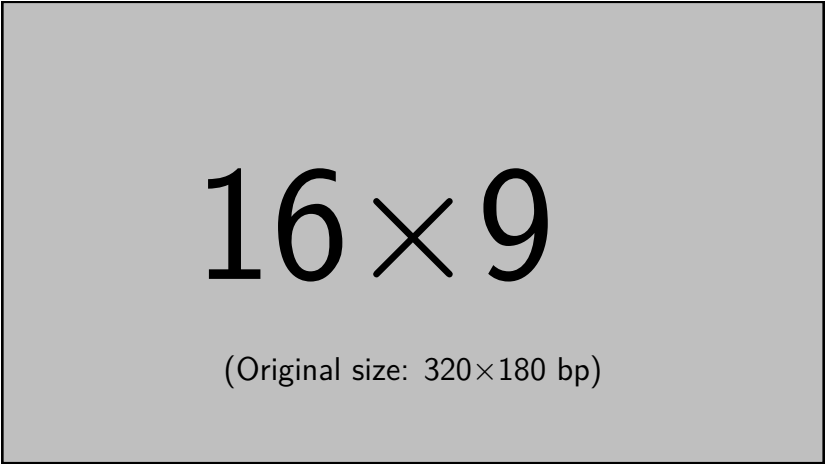
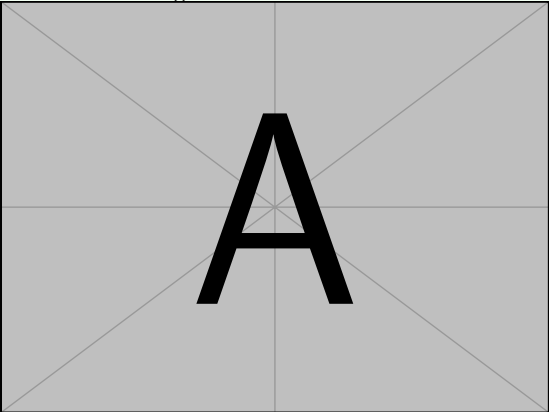


Figure 4. 16x9 Figure

Corresponding L^AT_EX code of ./paper.tex

803 æ£üŸòŰ
804 % Source:
805 <https://tex.stackexchange.com/a/468994/9075>
806 \begin{figure}
807 \caption{Table with diagonal line}
808 \label{tab:diag}
809
810
811
812

814
815
816 \end{center}
817 \end{figure}



(a) Case I

K. Source Code

Listing 1 shows source code written in XML. Line 2 contains a comment.

1 <listing name="example">

(b) Case II

Figure 5. Example figure with two sub figures.

(a)

(b)

Figure 6. Example figure with two sub figures. IEEE style. (a) The first case. (b) The second case.

Figure 7. Simple Table

Heading1	Heading2
One	Two
Thee	Four

Figure 8. Table with diagonal line

Diag Column Head I	Diag Column Head II	Second	Third
		foo	bar

```

2 <!-- comment -->
3 <content>not interesting</content>
4 </listing>

```

Listing 1. Example XML Listing

Corresponding L^AT_EX code of ./paper.tex

```

823 %\begin{lstlisting}
824 \Cref{lst:XML} shows source code written in XML.
825 \Cref{line:comment} contains a comment.
826
827 \begin{lstlisting}[
828   language=XML,
829   caption={Example XML Listing},
830   label={lst:XML}]
831 <listing name="example">
832 <!-- comment --> (* \label{line:comment} *)
833 <content>not interesting</content>
834 </listing>
835 \end{lstlisting}

```

One can also add float as parameter to have the listing floating. Listing 2 shows the floating listing.

Corresponding L^AT_EX code of ./paper.tex

```

841 %\begin{lstlisting}
842 \begin{lstlisting}[
843   % one can adjust spacing here if required
844   % aboveskip=2.5\baselineskip,
845   % belowskip=-.8\baselineskip,
846   float,
847   language=XML,
848   caption={Example XML listing -- placed as
849     floating figure},
849   label={lst:flXML}]
850 <listing name="example">
851   Floating
852 </listing>
853 \end{lstlisting}

```

```

1 <listing name="example">
2   Floating
3 </listing>

```

Listing 2. Example XML listing – placed as floating figure

```

1 {
2   key: "value"
3 }

```

Listing 3. Example JSON listing – placed as floating figure

```

1 public class Hello {
2     public static void main (String[] args) {
3         System.out.println("Hello World!");
4     }
5 }

```

Listing 4. Example Java listing

One can also typeset JSON as shown in Listing 3.

Corresponding L^AT_EX code of ./paper.tex

```

858 %\begin{lstlisting}
859 \begin{lstlisting}[
860   float,
861   language=json,
862   caption={Example JSON listing -- placed as
863     floating figure},
863   label={lst:json}]
864 {
865   key: "value"
866 }
867 \end{lstlisting}

```

Java is also possible as shown in Listing 4.

Corresponding L^AT_EX code of ./paper.tex

```

872 %\begin{lstlisting}
873 \begin{lstlisting}[
874   caption={Example Java listing},
875   label=lst:java,
876   language=Java,
877   float]
878 public class Hello {
879     public static void main (String[] args) {
880         System.out.println("Hello World!");
881     }
882 }
883 \end{lstlisting}

```

L. Itemization

One can list items as follows:

- Item One
- Item Two

Corresponding L^AT_EX code of ./paper.tex

```

890 %\begin{itemize}
891 \begin{itemize}
892   \item Item One
893   \item Item Two
894 \end{itemize}

```

With the package paralist, one can create itemizations with lesser spacing:

- Item One

- Item Two

Corresponding L^AT_EX code of ./paper.tex

```
899 %EÜÜöÜ
900 \begin{compactitem}
901 \item Item One
902 \item Item Two
903 \end{compactitem}
```

One can enumerate items as follows:

- 1) Item One
- 2) Item Two

Corresponding L^AT_EX code of ./paper.tex

```
908 %EÜÜöÜ
909 \begin{enumerate}
910 \item Item One
911 \item Item Two
912 \end{enumerate}
```

With the package paralist, one can create enumerations with lesser spacing:

- 1) Item One
- 2) Item Two

Corresponding L^AT_EX code of ./paper.tex

```
917 %EÜÜöÜ
918 \begin{compactenum}
919 \item Item One
920 \item Item Two
921 \end{compactenum}
```

With paralist, one can even have all items typeset after each other and have them clean in the TeX document:

- 1) All these items... 2) ...appear in one line 3) This is enabled by the paralist package.

Corresponding L^AT_EX code of ./paper.tex

```
926 %EÜÜöÜ
927 \begin{inparaenum}
928 \item All these items...
929 \item ...appear in one line
930 \item This is enabled by the paralist package.
931 \end{inparaenum}
```

M. Other Features

The words “workflow” and “dwarflike” can be copied from the PDF and pasted to a text file.

Corresponding L^AT_EX code of ./paper.tex

```
936 %EÜÜöÜ
937 The words \enquote{workflow} and
\enquote{dwarflike} can be copied from the
PDF and pasted to a text file.
```

The symbol for powerset is now correct: $\wp$ and not a Weierstrass p ($\wp$).

$\wp(1, 2, 3)$

Corresponding L^AT_EX code of ./paper.tex

```
940 %EÜÜöÜ
941 The symbol for powerset is now correct:
\powerset$ and not a Weierstrass p ($\wp$).
942
943 $\powerset(\{1,2,3\})$
```

Brackets work as designed: <test> One can also input backticks in verbatim text: `test`.

Corresponding L^AT_EX code of ./paper.tex

```
946 %EÜÜöÜ
947 Brackets work as designed:
<test>
One can also input backticks in verbatim text:
\verb|`test`|.
```

IV. CONCLUSION AND OUTLOOK

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

ACKNOWLEDGMENT

Identification of funding sources and other support, and thanks to individuals and groups that assisted in the research and the preparation of the work should be included in an acknowledgment section, which is placed just before the reference section in your document [4].

REFERENCES

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- [3] M. Scharrer, *The mwe Package*, 2017. [Online]. Available: <http://texdoc.net/mwe>
- [4] B. Veytsman, “Latex class for the association for computing machinery – acknowledgement information,” Aug. 2021. [Online]. Available: <https://github.com/borisveytsman/acmart/blob/1704c8bf7eee92a1515ff755f5118b6a22bb1f8e/samples/samples.dtx#L709>

All links were last followed on October 5, 2020.